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Impact of climate change on Extreme Sea Levels

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CONTENTS

1 What is the driver of coastal flooding?

2 including the climate change impacts

3 Methodology details

4 Regional workshops

5 Towards application

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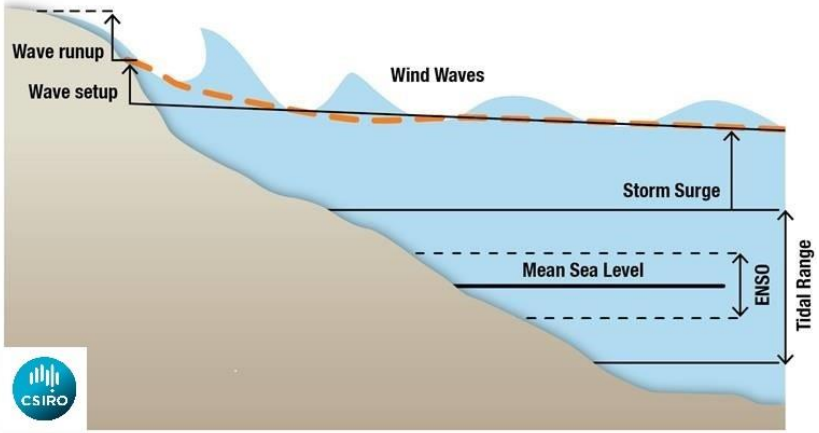
[Credit: USA TODAY NEWS]

Extreme sea level

tide + storm surge + wave setup + sea level rise



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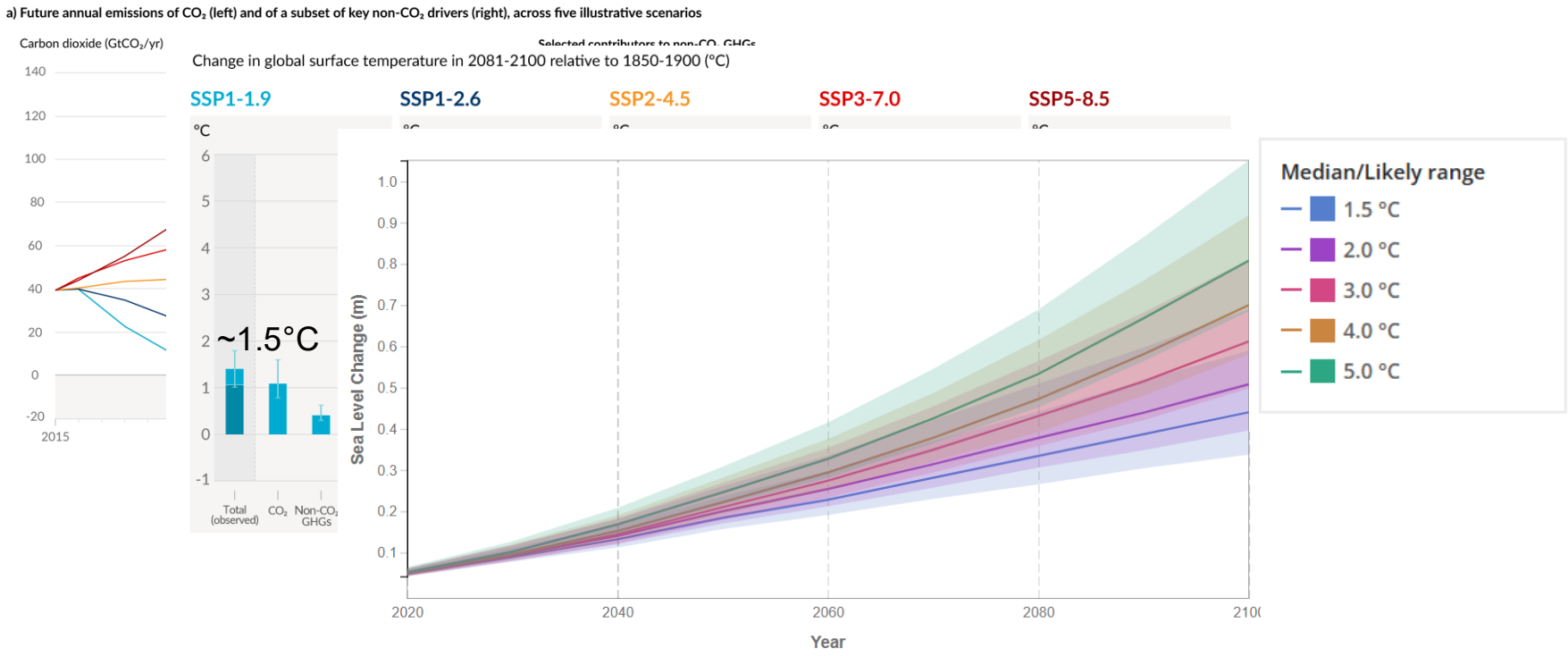
RELATIVE SEA LEVEL RISE

Relative sea level rise contributes to increases in the frequency and severity of coastal flooding in low-lying areas and to coastal erosion along most sandy coasts (*high confidence*).

sea level rise by warming levels



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PUBLICATION AFTER AR6 WGI

In which part of the world a 100-year event will becoming an annual event if global warming reached 1.5°C – 5°C by the end of 21st century?

ARTICLES
https://doi.org/10.1038/s41558-021-01127-1
nature climate change
Check for updates

OPEN

Extreme sea levels at different global warming levels

Claudia Tebaldi[1]✉, Roshanka Ranasinghe[2,3,4], Michalis Vousdoukas[5], D. J. Rasmussen[6], Ben Vega-Westhoff[7], Ebru Kirezci[8], Robert E. Kopp[9], Ryan Sriver[7] and Lorenzo Mentaschi[5,10]

The Paris agreement focused global climate mitigation policy on limiting global warming to 1.5 or 2 °C above pre-industrial levels. Consequently, projections of hazards and risk are increasingly framed in terms of global warming levels rather than emission scenarios. Here, we use a multimethod approach to describe changes in extreme sea levels driven by changes in mean sea level associated with a wide range of global warming levels, from 1.5 to 5 °C, and for a large number of locations, providing uniform coverage over most of the world's coastlines. We estimate that by 2100 ~50% of the 7,000+ locations considered will experience the present-day 100-yr extreme-sea-level event at least once a year, even under 1.5 °C of warming, and often well before the end of the century. The tropics appear more sensitive than the Northern high latitudes, where some locations do not see this frequency change even for the highest global warming levels.

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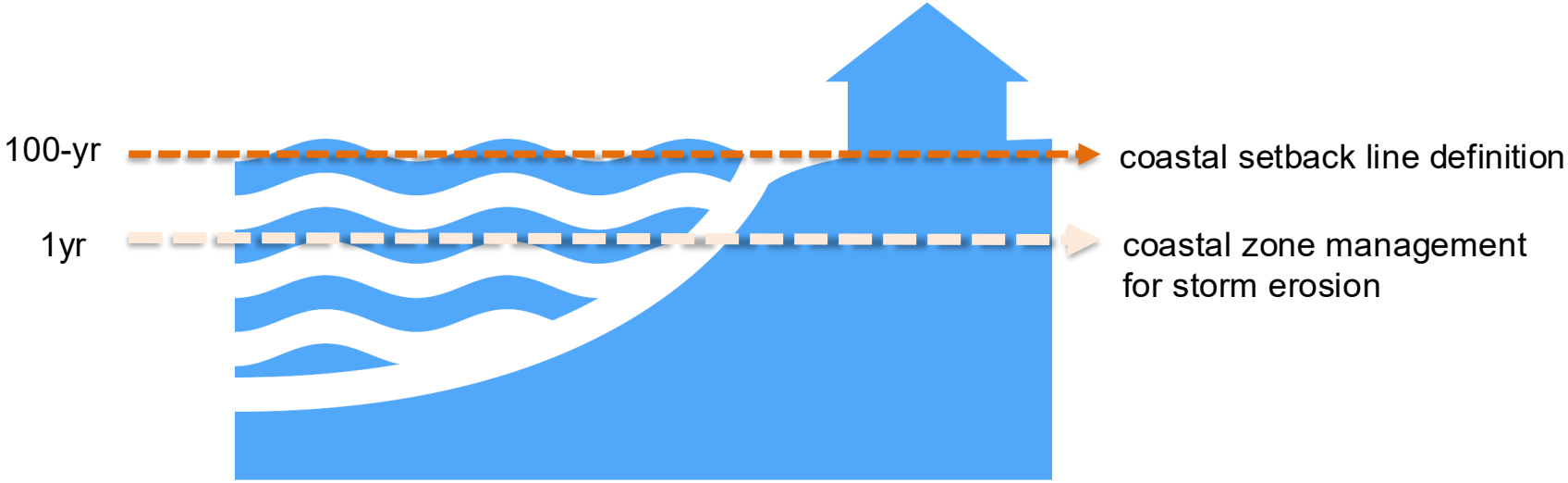
UNDER WHICH WARMING LEVEL?



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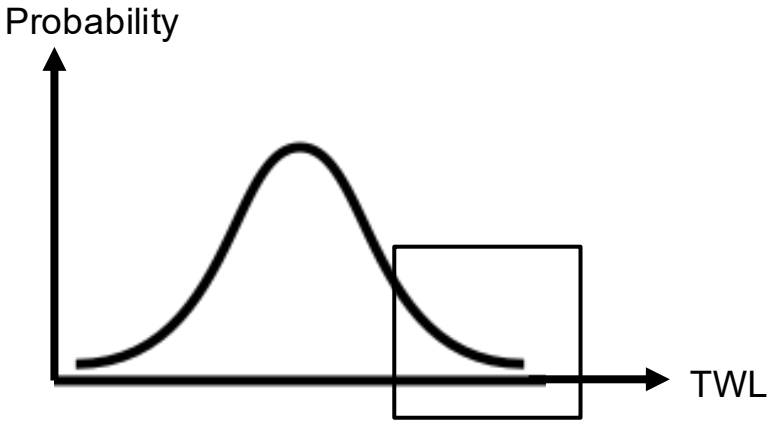

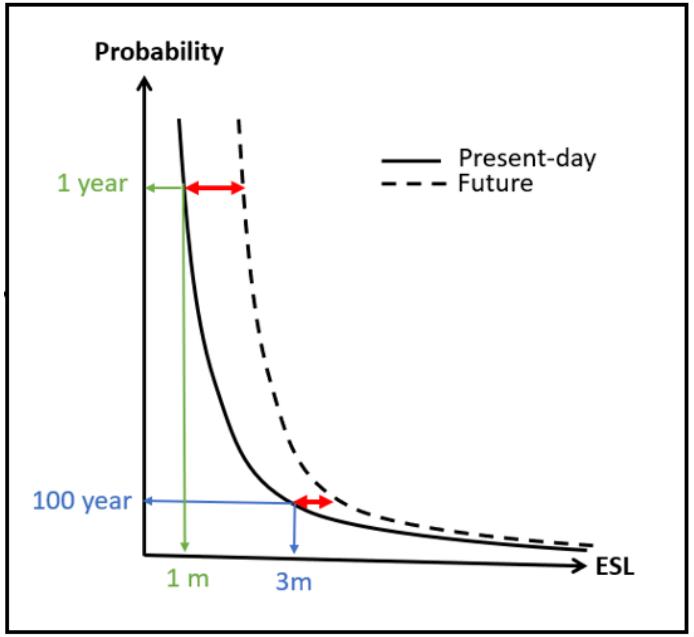

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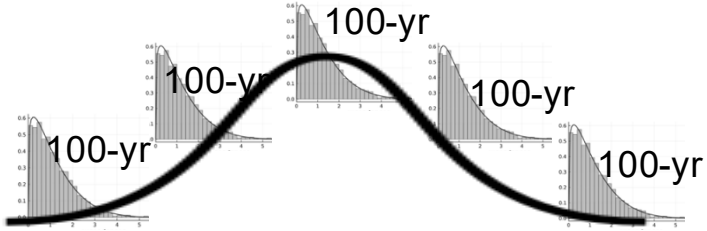
Integrating uncertainty

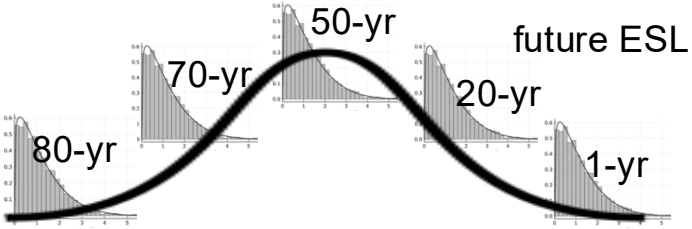
present-day ESL



+

randomly picked SLR (x°C)

=



future ESL under x°C

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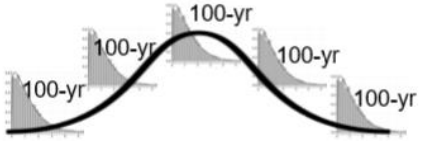
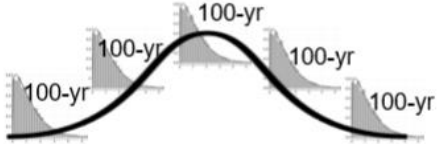
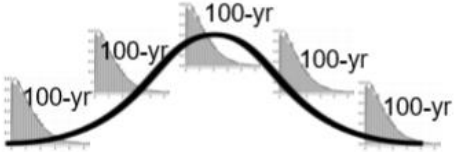
3 present-day ESL datasets



Vousdoukas et al. (2018)



Kireczi et al. (2018)



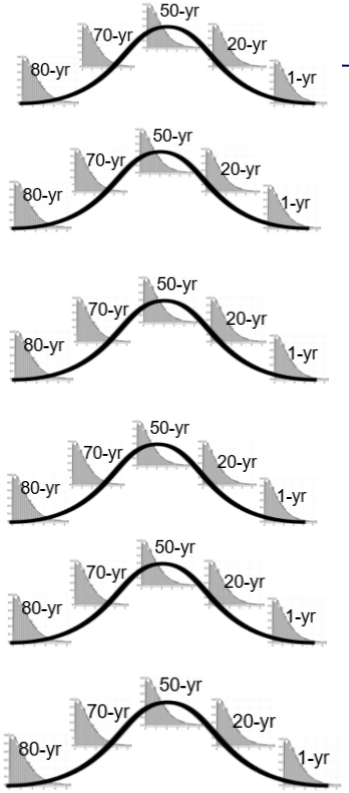
Rasmussen et al. (2018)

2 sea level rise projections for x°C

Rasmussen et al. (2018)

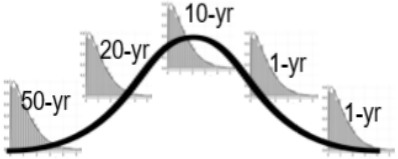
Vega-Westhoff et al. (2019)

6 alternatively paired ESL projections for x°C



ESL full convolution for x°C

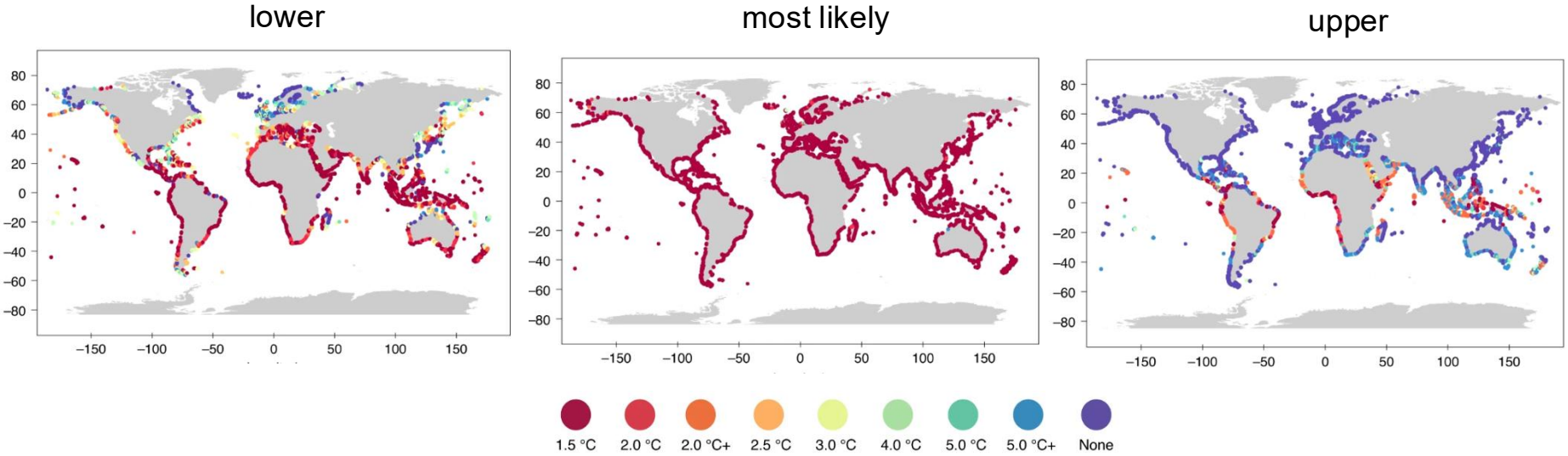
most likely



best-case

worst-case

Present-day 100-yr ESL event to become at least an annual event – under which GWL?



"~50% of the locations considered will experience the present-day 100-yr extreme-sea-level event at least once a year, even under 1.5 °C of warming"

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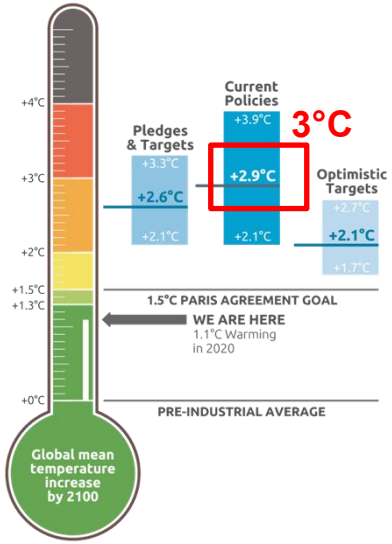
FOR CLIMATE RESILIENT COASTLINES, WE NEED



Climate Action Tracker

CAT warming projections
Global temperature increase by 2100

December 2020 Update

Projections for different global warming levels and return periods

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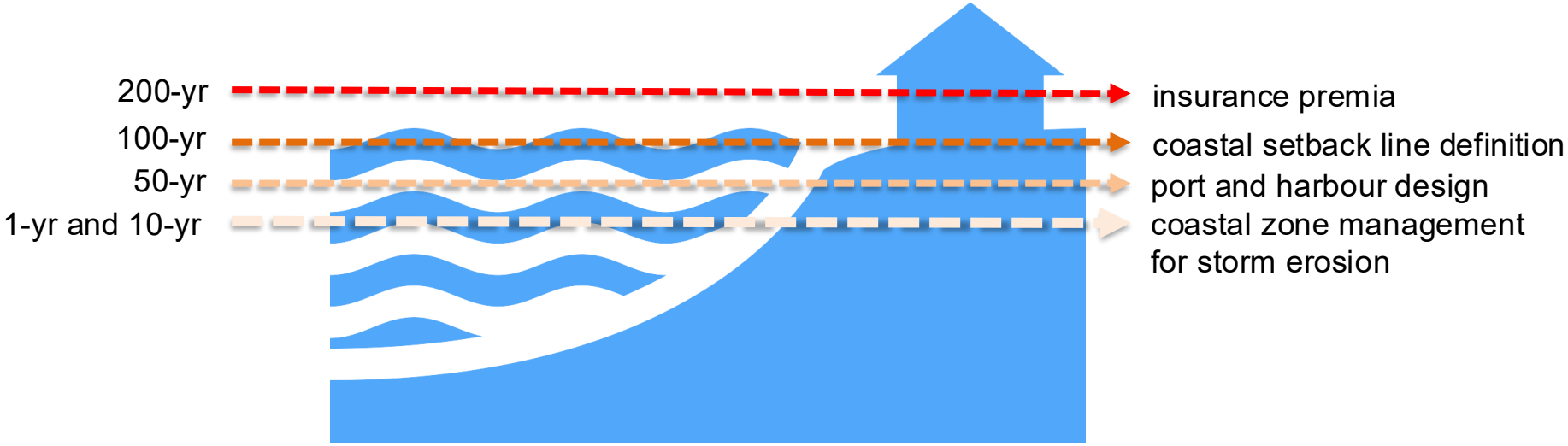
DIFFERENT RETURN PERIODS



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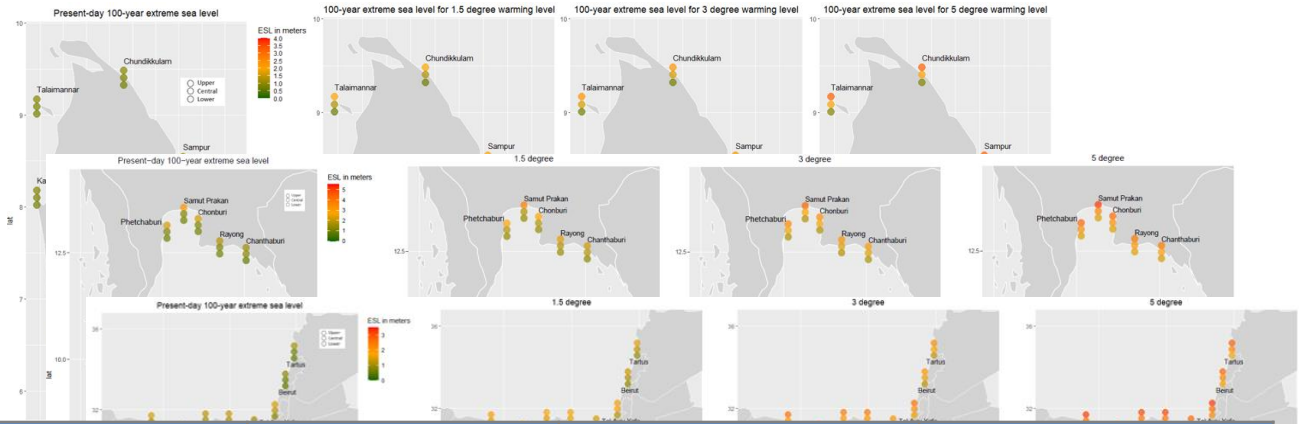
GWL IMPACT ON OTHER RPS –SRI-LANKA, THAILAND, EGYPT



Locations	+1.5° C	+3° C	+5° C
El-Alamein	2080 [2040 - >2100]	2070 [2040 - >2100]	2060 [2030 - 2100]
Alexandria	2080 [2040 - >2100]	2070 [2040 - >2100]	2060 [2030 - 2090]
Kafr-El-Sheikh	2080 [2040 - >2100]	2070 [2040 - >2100]	2060 [2040 - 2100]
Damietta	2080 [2040 - >2100]	2070 [2040 - >2100]	2060 [2040 - >2100]
Port-Said	2090 [2040 - >2100]	2070 [2040 - >2100]	2060 [2030 - >2100]

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GWL IMPACT ON OTHER RPS – EGYPT, THAILAND, SRI-LANKA

- ESLs around all these countries are projected to rise significantly, with potential increases ranging from two-fold to several times the current levels.
- ESL increase is higher for 1 yr and 10 yr RP ESLs, and there is not that much difference among the increases (%) for other RP ESLs
- The increase in ESLs becomes highly pronounced for GWL > 3°C.
- Achieving Paris agreement temperature target will have more adaptation time for the Nile Delta

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APPLICATION TOWARDS CLIMATE RESILIENT COASTLINE

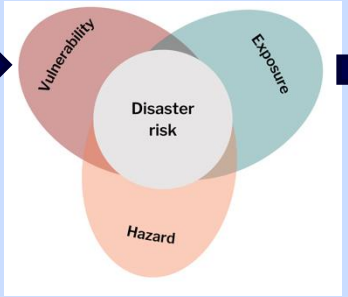
CC impact in the driver

projections of ESL based on hydrodynamic modelling or observed data

Climate change integrated coastal flood maps

coastal flood maps forced by ESL projections and based on high resolution elevation models (Fully dynamic coastal flood modeling (SFINCS))

coastal flood risk assessment



Adaptation measures and future planning

coastal flood defenses, relocate, retreat, accommodate, etc.

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THANK YOU

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